

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 899706156 | hariramakrrish@gmail.com | [LinkedIn](#) | Stamp 1G — Eligible to work full-time in Ireland

PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Designed and deployed **machine learning pipelines** using **Python (scikit-learn, pandas, NumPy)** and **SQL** to operationalize predictive models across enterprise healthcare data, achieving **92% accuracy** on classification tasks and reducing inference latency by **40%** through model optimization and **AWS Lambda** containerization.
- Engineered **end-to-end data science workflows** leveraging **Power BI** and **Tableau** for real-time dashboard development, **DAX** measure design, and **Power Query** ETL pipelines; delivered **15+ interactive executive dashboards** supporting clinical and operational decision-making for **200+ stakeholders**.
- Developed and maintained **production AI/ML infrastructure** using **Docker**, **Jenkins CI/CD pipelines**, and **AWS (EC2, S3, CloudWatch)** to automate model training, validation, and deployment; reduced time-to-production for new models by **55%** and achieved **99.8% SLA uptime** for inference services.
- Conducted **statistical analysis** and **hypothesis testing** using **R** and **RStudio** on large-scale healthcare datasets; performed **time-series forecasting**, **regression analysis**, and **A/B testing** across clinical and business processes, identifying **12 actionable insights** that informed strategic initiatives and improved patient outcomes.
- Led **data governance** and **compliance frameworks** for healthcare AI systems, ensuring adherence to **HIPAA**, **GDPR**, and **FDA AI/ML validation requirements**; designed audit trails, data lineage documentation, and model explainability protocols supporting regulatory submissions and cross-functional audit teams.
- Deployed **advanced analytics** and **machine learning models** for healthcare operations including **predictive analytics** for patient readmission, **anomaly detection** for clinical outliers, and **unsupervised learning** for patient segmentation; integrated models with **REST APIs** and **enterprise ERP systems** enabling real-time clinical decision support across **50+ hospitals**.

SKILLS

Machine Learning & AI – Supervised learning, unsupervised learning, regression, classification, clustering, ensemble methods, neural networks, model evaluation metrics, hyperparameter tuning, scikit-learn, TensorFlow, PyTorch.

Data Science & Analytics – Predictive analytics, statistical analysis, hypothesis testing, A/B testing, time-series forecasting, exploratory data analysis, feature engineering, anomaly detection, patient segmentation.

Data Engineering & Pipelines – ETL/ELT pipeline development, data warehousing, SQL optimization, data modelling, Apache Spark, Apache Hadoop, data validation, data quality frameworks.

Business Intelligence & Visualization – Power BI, Tableau, DAX, Power Query, interactive dashboards, data storytelling, executive reporting, KPI design, BI architecture.

Programming & Big Data – Python, R, Java, SQL (PostgreSQL, MySQL, Oracle), pandas, NumPy, scikit-learn, Spark, Hadoop, shell scripting, Linux.

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Jenkins, CI/CD pipelines, Git, containerization, infrastructure as code, model deployment automation.

Healthcare & Compliance – HIPAA, GDPR, FDA AI/ML validation, data governance, audit trails, explainability (XAI), regulatory documentation, clinical data standards, healthcare AI frameworks.

EDUCATION AND TRAINING

PROJECTS

Real-Time Predictive Analytics & AI Model Operations Platform (HCL Technologies)

- Architected and deployed a **production-grade machine learning platform** using **Python (scikit-learn, TensorFlow)**, **Spark**, and **AWS Lambda** to operationalize **12+ predictive AI models** across healthcare enterprise data; achieved **94% model accuracy** and **<2-second inference latency** at scale, processing **500K+ daily predictions**.
- Engineered **end-to-end data pipelines** using **SQL (PostgreSQL, Oracle)**, **Power Query**, and **Spark ETL** to ingest, transform, and validate healthcare datasets from **50+ operational systems**; implemented **data quality frameworks** reducing data errors by **87%** and enabling real-time model retraining every **6 hours**.
- Designed **Power BI** and **Tableau executive dashboards** with **DAX** measures for model performance monitoring, KPI tracking, and clinical outcome visualization; delivered **8 published dashboards** to **150+ healthcare stakeholders**, enabling data-driven clinical decision-making.
- Implemented **HIPAA-** and **GDPR-compliant data governance and model validation frameworks** including automated audit trails, feature lineage tracking, and **explainability (XAI)** protocols; ensured **FDA AI/ML readiness** through comprehensive documentation supporting regulatory submissions.
- Built **CI/CD automation** using **Jenkins**, **Docker**, and **AWS** infrastructure to orchestrate model training, validation, and production deployment; reduced time-to-production for new AI models by **58%** and achieved **99.9% inference service availability**.

Healthcare Operations Intelligence & Statistical Analytics Suite (HCL Technologies)

- Conducted **statistical analysis** and **A/B testing** using **R**, **RStudio**, and advanced **hypothesis testing** methodologies to evaluate clinical interventions and operational improvements; generated **18 peer-reviewed statistical reports** informing strategic initiatives across **25+ clinical departments**.
- Developed **predictive analytics models** including **time-series forecasting**, **regression analysis**, and **patient segmentation** using **unsupervised learning** techniques; deployed segmentation models achieving **89% silhouette score**, enabling personalized clinical pathways for **100K+ patients**.
- Integrated **machine learning models** with **REST APIs** and **enterprise ERP systems** to operationalize analytics insights; connected **6 predictive models** to downstream operational workflows, reducing manual clinical review time by **65%** and enabling real-time clinical decision support.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)